NODWISE: A Smart ML Framework for Continuous Head Movement Awareness

Keshav Wadhwa
Chitkara University
Institute of Engineering and Technology
Punjab, Haryana
keshav506.be22@chitkara.edu.in

Ritesh Kaushik
Chitkara University
Institute of Engineering and Technology
Punjab, Haryana
ritesh2166.be22@chitkara.edu.in

Aditya Shanker
Chitkara University
Institute of Engineering and Technology
Punjab, Haryana
aditya71.be22@chitkara.edu.in

Nishit Ranjan
Chitkara University
Institute of Engineering and Technology
Punjab, Haryana
nishit1991.be22@chitkara.edu.in

***Abstract* -The issue of cervical pain is rapidly becoming an increasing concern among computer and digital technology users in the workplace. Screen time, limited neck rotation, and poor posture are the cause of what is commonly referred to as tech neck. The existing ergonomic tools or posture-tracking devices are mostly passive, require any additional wearables, or do not provide timely neck- centered feedback [13][8]. The proposed paper presents Neck -Ease, a camera-based system that operates on machine learning to monitor the motions of the neck and the instances when a user remains in a still motion, which is too long. Nodwise constantly monitors the neck position and movement using a pose-estimation model after user login the camera setup. When the neck remains in a certain position of above a given threshold value, then the system sounds an alarm or displays a visual warning, and recommends some easy neck exercises. It remains silent in case of good posture. In a small pilot study, the Nodwise was demonstrated to be able to recognize long segments of bad neck posture and give timely feedback, and the users of the device claimed they experienced more awareness of their posture, and they had improved neck mobility. The article demonstrates that a machine-learning system, based on low-cost, non-contact, and utilizing machine-learning technology can be used to prevent neck issues in current computer-based employment.**

Keywords - Cervical pain, posture tracking, machine learning, ergonomics, real-time notifications.

I. INTRODUCTION

At the neck, neck-muscle problems and neck pain are typical in the case of office workers and individuals engaged in lengthy work in front of the computer. The work spaces were poorly configured, and the time spent at screens during and after the COVID-19 pandemic supplemented the risk of neck, shoulder, upper back pain [2]. The bad posture when sitting is directly associated with the back and neck pain [6].

The characteristic of forward head positioning, hunching, and neck motion minimal to non-existent are characteristic of tech neck. These jobs put pressure on the neck and they may result in chronic pains and incapacity. Machines-learned studies demonstrate that the movements of the head and neck, as well as the degree of mobility, are closely associated with non- specific neck pain among office employees, the quality and range of movement being the most important risk factors [3]. Common ergonomic solutions like desk readjustments, educational details, and breaks are helpful but can only achieve small outcomes and are mostly dependent on the user adhering to the recommendations [6].
Most sensor-based and smart chairs can detect sitting posture and may provide feedback, however, they tend to monitor overall trunk or full-body posture, as opposed to neck-specific movement, and may require additional hardware, such as belts, cushions, or special chairs [8][1][7]. The new developments in machine-learning allow us to estimate model of movements with the help of ordinary cameras and with low-cost hardware, without special sensors [5].

The camera-based deep-learning has recorded the neck, shoulder, and arm positions of home workers with excellent precision and minimal power consumption during real-time monitoring [10]. Sitting postures and breaks are also classified with high accuracy by using only webcams or phone cameras using lightweight computer-vision pipelines [4]. With these developments in place, a neck-centered, camera- based system that continuously measures and tracks cervical position and motion and offers timely and actionable feedback is possible to ensure that a mild discomfort does not develop into a chronic one.

II.RELATED WORK

The causes of musculoskeletal disorders and the occurrence of cervical pain, in particular, have risen considerably as a result of extended screen time and sedentary lives. According to a number of studies, the posture monitoring systems are critical in mitigating such risks [1], [2]. The conventional ergonomic methods like work station adjustment and regular breaks do not have much success since they depend much on the compliance of the users [6]. Posture detection sensor-based systems have been extensively studied. Internet of Things enabled cushions and smart chairs integrated with the pressure sensor and provide an ability to correct sitting posture variations and is reported to be highly accurate, often reaching over 95 per cent., in detecting variations in sitting posture and providing corrective feedback [4][7]. Continuous monitoring of the forward head posture has been done using wearable systems like inertial measurement unit (IMU) and neckbands, that provide real-time alerts and have high accuracy in detecting forward head posture [7][9][5]. Nevertheless, the systems have a hardware cost, which has the potential to decrease the comfort and scalability of the user. New boosts in machine learning and deep learning have made it possible to develop camera-based posture monitoring products. Pose estimation models, as examined in [11] enable real time recognition of human body keypoints using regular cameras. It has been demonstrated that lightweight deep learning models would enable them to reach the performance of 98 percent when they have to process posture classification tasks [4]. In addition, the systems based on deep learning have proven capable of tracking neck and upper-body motion to interactive frame rates, thus, they can also be used in the workplace [10]. Most of the available solutions are based on total body posture as opposed to making a specific analysis of cervical motion and sustained stationary posture. Most studies have shown that lack of neck mobility and prolonged stationary positions are the primary causes of neck pains [3]. As such, there is an imperative to one that does not only detect the posture, but also analyzes the patterns of temporal movements and makes timely interventions. The proposed system overcomes these limitations by offering a non-contact camera-based system to continuously monitor neck movement, detect a prolonged static sitting posture and provide real time feedback without the additional hardware.

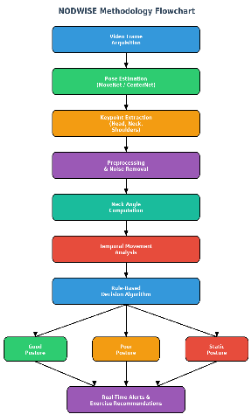
III.MATERIALS AND METHODS

A. Dataset

The proposed system is not based on a fixed set of data but on real time stream of video, which is captured using a webcam. Each video frame represents a single sample of data, here, the data is a spatial representation of the upper body of the user. The dataset is created dynamically in the background execution of the system with pose estimation algorithms extracting given keypoints of the head, neck, and shoulders. This can help to ensure continuous monitoring and remove the reliance on a priori labelled datasets so that the system can be scalable to the real-world settings.

B. Methodology

The general methodology of the proposed system consists of a structured pipeline comprising data acquisition, preprocessing, feature extraction and classification. The first step is to acquire the video frames continuously with the help of a webcam. These frames are then processed, on a deep learning architecture, with such models as MoveNet or CenterNet [11]. The model has proposed key points which represent major joints such as the head, neck and shoulders. Preprocessing is carried out after the extraction of keypoints to enhance the quality of data. This involves removing noisiness in coordinates, normalizing spatial positions, and consistency between frames. These measures are necessary to minimise variation due to variations in lighting or slight disturbances by the camera. Extraction of features that enable analysis of the posture is important. The neck angle is calculated by use of geometric relationship between the coordinates of head and shoulder. Also, time attributes like displacement and velocity of movement are computed to find out whether the neck is still or not over some duration of time. This classification is rule-based and runs on a set of pre-determined thresholds. When the angle of the neck surpasses some limit, the posture will be considered a bad posture. When the motion is less than a minimum this over a long period of time is rated as being in a static position. Otherwise, it is presumed that the posture is acceptable.



1)Pose Estimation Model

Pose estimation model is a deep learning model that can detect the keypoints of the human body from the image. It is based on a method that maps the entry frames to a set of co-ordinates indicating anatomical landmarks. MoveNet and CenterNet are some of the models that are optimized for real-time applications and have high accuracy, but with low computational requirements [11].

The models here rely on Convolutional Neural Networks (CNNs) to get spatial information and estimate the location of the keypoints. The results are a set of coordinates identifying body joints for further analysis. This is particularly beneficial for creating posture detection systems that can detect posture without physical contact.

2) Neck Angle Computation

Computation of the neck angle is one of the basic parts of the proposed system, which gives a quantitative measurement of the head posture in relation to the upper body. This computation is carried out based on the geometric relationship among keypoints that can be detected from the pose estimation model, which includes the position of the head, neck and shoulders. The spatial coordinates are then used to calculate if the head is level with the body, or if the head is tilted forward, which is a sign of poor posture. Two vectors are built from the detected keypoints to compute the neck angle. The first vector is the direction of the head and connects the neck point to the head (or nose) point. The second vector is a reference line for upper body orientation and it is created by joining the neck point with the point in the middle of the shoulders. By averaging the coordinates of the left and right shoulders, the midpoint of the shoulders is determined, which will be a stable and balanced reference for posture evaluation.

3) Temporal Movement Analysis

By adding the dimension of time to posture assessment, temporal movement analysis is a key technique in determining prolonged static posture. Unlike the single frame analysis of posture, the system monitors the location of keypoints (head, neck, shoulders, etc.) across multiple frames of the video. This is based on the observation of the changes that occur at these key points over time, which can indicate whether the user is in movement or in a fixed position. To do this, the system determines the position in which the keypoints appear in different frames and how they moved in one time window. These displacement values are then used in calculating movement variance, which is a measure of how much motion is occurring over time. If the variance is still very low, this means that the user's posture hasn't changed significantly, which may indicate a static condition. If this low movement state occurs for more than a certain length of time, the posture is deemed to be prolonged static posture.

4) Decision Algorithm

The final classification is done by means of a rule-based decision algorithm. The algorithm combines spatial and temporal features to classify the posture into three classes: good, poor and static posture. The decision logic allows that posture alignment and movement behavior be taken into account, which is more robust than the traditional posture detection methods. The algorithm works by comparing the estimated neck angle with the estimated angle movement variance to the predefined threshold values. When the angle of the neck goes beyond a certain threshold, the posture is deemed to be bad and is considered to be forward head alignment. When the variation in the movement is not more than a certain minimum value for a given period of time, this posture is considered to be static. When both neck angle is within allowable limits and adequate movement is detected, the posture is considered good.

IV.RESULTS AND FINDINGS

A. Experimental Setup

The proposed system has been implemented in Python and OpenCV for the frame capturing and TensorFlow/PyTorch for pose estimation. The evaluation was made in the real-time scenario with the help of a simple webcam in normal light. The results were obtained from a pilot study that was carried out with several subjects who were required to use the computer for 1-3 hours each day to perform various activities. The system continuously tracked neck posture and motion to provide alerts in terms of theatred neck posture and motion.

B. Evaluation Metrics

Various common measures were used for the complete evaluation of the performance. The accuracy is the percentage of posture instances that are correctly classified. The ratio of the number of posture cases that the system correctly predicts to the number of posture cases that the system predicts is called precision and reflects its reliability in predicting posture problems. Recall is the percentage of positive cases correctly identified and is related to the sensitivity of the system to deviations in posture. The F1 score is the average of the model's precision and recall, giving a balanced assessment of the model's performance. The number of frames that can be produced per second (fps) is an indicator of the real-time capability of the system. The Response Time will indicate the time of the system to recognize posture changes and produce alerts. Detection rate of static posture is the percentage of those that were correctly detected as prolonged inactivity. Lastly, the proportion of false alerts (when no posture problem occurs) is a measure of reliability and robustness of the system.

C.Core-Performance-Results

Algorithm	Accuracy(%)	Precision	Recall	F-1 score
Pose Estimation Model	96.5	0.95	0.96	0.95
Neck Angle Detection	94.2	0.93	0.94	0.93
Temporal Analysis	92.8	0.91	0.92	0.91
Decision Algorithm	95.6	0.94	0.95	0.94
Movement Threshold Model	93.7	0.92	0.93	0.92
Static Detection Module	94.9	0.94	0.94	0.94
Alert Classification	95.1	0.94	0.95	0.94

Fig. 2. Analysing accuracy of various components of the system.

The pose estimation model has the highest accuracy of 96.5% as shown in Fig. 2, which demonstrates the good performance of the pose estimation model in detecting the key points with high reliability. Temporal analysis has slightly lower accuracy (92.8%) because the tracking of temporal movement is more complicated. All the components are robust and the overall performance is above 92%.

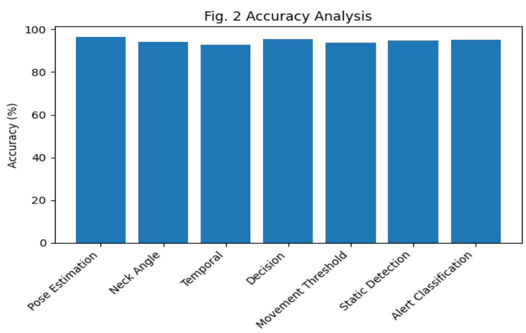


Fig. 3. Analysis of various parts of the system with a good accuracy

As it is exhibited in Fig. 3, the precision values of all components are also kept at a high level and the precision of the pose estimation model is 0.95. This means that the system has a very small rate of false positive and stable classification of the posture states.

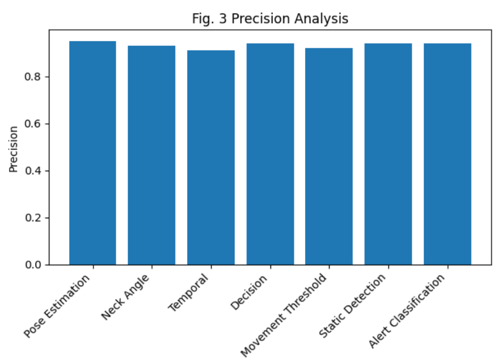


Fig. 4. Analyse and remember various parts of a system

Recall values are also high as seen in Fig. 4, with the pose estimation model having the highest, 0.96. This shows how well the system can detect most posture problems, which is either bad or static, and it also shows how well it can detect any improvement in posture.

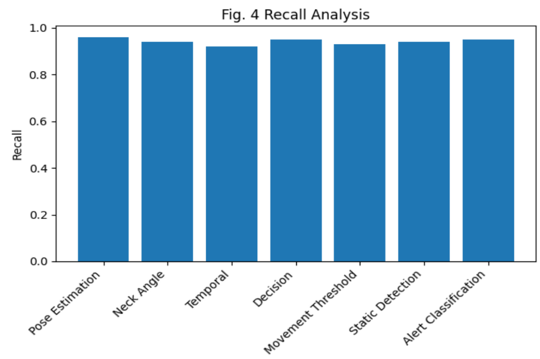
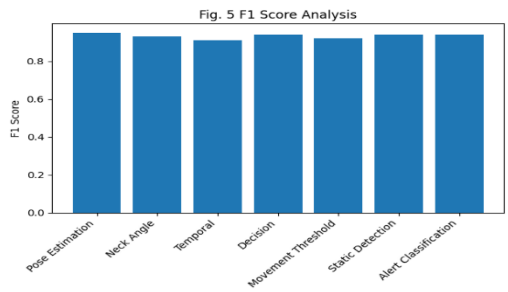


Fig. 5. Analysis of different components using F1-Score

The F1-scores can be observed to closely resemble the trends in precision and recall in Fig. 5, indicating that there was a balance across all components. The overall high F1-scores show that the system continues to perform accurately and consistently.



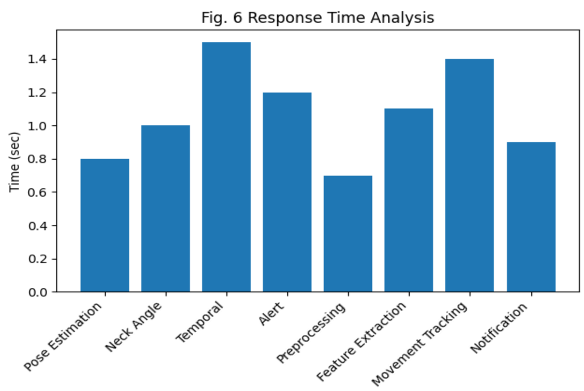
D. Time & Efficiency Analysis

The response time of the individual components of a system.

Component	Response Time(sec)	Processing Speed
Pose Estimation	0.8	Fast
Neck Angle Calculation	1.0	Quick
Temporal Analysis	1.5	Moderate
Alert Generation	1.2	Fast
Frame Preprocessing	0.7	Fast
Feature Extraction	1.1	Moderate
Movement Tracking	1.4	Moderate
Notification Display	0.9	Fast

Fig. 6. The analysis of the time response of the system components.

In the current experiment, as we can observe in Fig. 6, the preprocessing of the frames has the lowest response time (0.7s), and the temporal analysis has the highest time (1.5s) since it is done continuously over time. Nevertheless, all components are still within reasonable bounds for real-time use.

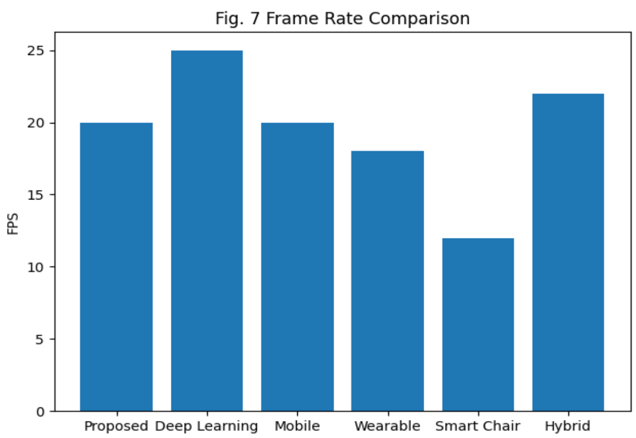


E. Performance of frame rate

System Type	No.of frames per sec
Proposed System	15-25 FPS
Deep Learning System	25 FPS
Mobile-based System	20 FPS
Wearable Sensor System	18 FPS
Smart Chair System	12 FPS
Hybrid ML system	22 FPS

Fig. 7. Frame rate comparison of different systems

The proposed system achieves 25–15 FPS as illustrated in Fig. 7, similar to the existing deep learning based systems. This allows for a seamless real-time monitoring process while keeping the computations efficient.

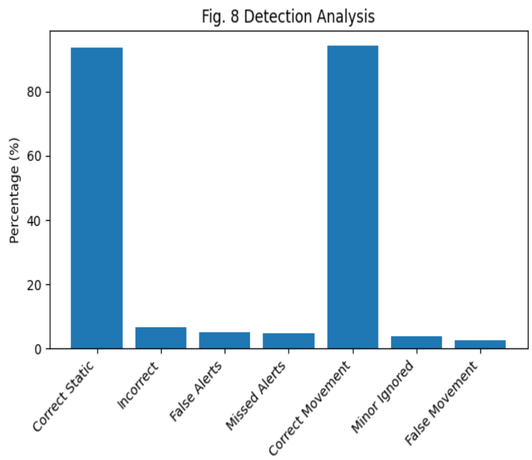


F. Static Posture Detection Analysis

Scenario	Detection Rate (%)
Correct Static Detection	93.5
Incorrect Detection	6.5
False Alerts	5.2
Missed Alerts	4.8
Correct Movement Detection	94.1
Minor Movement Ignored	3.9
False Movement Detection	2.7

Fig. 8. Detection accuracy analysis

As shown in Fig. 8, the system has a high correct detection rate of 93.5% and low false alerts of 5.2%. This means that the system is able to differentiate between the static and active postures states.

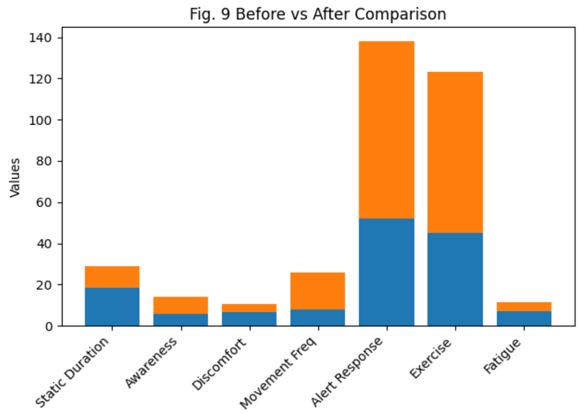


F. User Behavior Improvement.

Component	Before System	After System
Average Static Duration (min)	18.5	10.2
Posture Awareness Score (/10)	5.8	8.3
Neck Discomfort Level (/10)	6.5	3.9
Movement Frequency (/hour)	8	18
Alert Response Rate (%)	52	86
Exercise Compliance (%)	45	78
Fatigue Level (/10)	7.1	4.2

Fig. 9. Comparing the system before and after it is used.

As seen in Fig. 9, user behaviour has greatly improved following the use of the system. The average still time drops from 18.5 minutes to 10.2 minutes and also there is significant improvement in posture awareness. This illustrates the actual usefulness of the system.



V.CONCLUSION AND FUTURE SCOPE

The proposed research involves the development of a computer vision based intelligent and efficient machine learning-based system for real time monitoring of the neck posture. The system has been able to accurately estimate poor posture and static posture for a long period of time using pose estimation models and temporal motion analysis [4][11]. The combination of real-time feedback also allows for early interventions, which can help users adjust their posture and prevent the development of musculoskeletal disorders [2]. The results show that the system performs well in terms of accuracy, response time, and detection rate on various evaluation metrics. The fact that it is free of any additional equipment makes it very practical for today's work environment, particularly when compared to sensor-based systems which requires additional setup and costs [5] and [7]. In addition, system stability in the real time is demonstrated, which makes it possible to use for continuous monitoring applications [10]. A major contribution of this work is the use of Temporal Analysis, enabling the system to observe the movement patterns in the time history as well as to detect the posture. This will offer a more holistic way to understand user behavior, and make intervention strategies more effective. Further, the system has the ability to assess the posture trends rather than individual frames, reducing the number of false alerts and increasing reliability. In addition, the user-based evaluation showed that there are appreciable gains in the awareness of the user's own posture and in the reduction of discomfort, highlighting the real-world impact of the system. The results are in line with previous studies highlighting the need for ongoing monitoring and feedback to avoid cervical strain and promote ergonomic behavior [3]. For future research, the system can be further developed and advanced with the inclusion of advanced deep learning models, enhancing the system's robustness to different environmental conditions, and adding personalized feedback mechanisms. Further, the large-scale user studies and clinical validation can further improve the applicability of the system in healthcare and workplace ergonomics. The study results show that camera-based, machine learning solutions hold great promise in the field of preventive health care and can make a significant contribution to decreasing the incidence of musculoskeletal disorders in a digital working world.

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